

# Collins Culbert

## Electrical Engineering Student | Embedded Systems & Robotics Control

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### SUMMARY

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Junior Electrical Systems Engineering Technology student with hands-on experience across closed-loop robotics control, FPGA digital logic, and register-level embedded programming. Led the engineering build of a 10-servo, PID-controlled robotic hand that earned a Best Presentation award and finalist standing. Comfortable working end to end: schematic to firmware to live demo.

### TECHNICAL SKILLS

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|-----------------------------|-----------------------------------------------------------------------------------------|
| <b>Programming</b>          | Python, C/C++ (STM32 bare metal), MicroPython, JavaScript, HTML/CSS                     |
| <b>Hardware &amp; Tools</b> | Keil uVision, Quartus Prime (FPGA/DE10), KiCad, oscilloscopes, multimeters, DC motor    |
| <b>Generative AI</b>        | Anthropic Claude API integration, prompt engineering, AI-assisted workflow automation   |
| <b>Specializations</b>      | Closed-loop PID control, I <sup>2</sup> C sensor integration, FPGA state machine design |
| <b>Software</b>             | MS Office Suite, HubSpot (basics)                                                       |

### PROFESSIONAL EXPERIENCE

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#### Engineering Participant | GE Aerospace Electrical Engineering Simulation | Dallas, TX · 05/2025

- Analyzed avionics electrical systems using structured problem-solving and critical-thinking frameworks.
- Developed electrical system design plans and produced supporting technical documentation for proposed designs.
- Contributed to verification and validation planning to support design accuracy and reliability.

### PROJECTS

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#### Robotic Hand — Closed-Loop PID Control · MicroPython, RP2040/ESP32, AS5600, PCA9548A, I<sup>2</sup>C

- Co-designed and built a 10-servo DC motor-driven robotic hand (2 continuous-rotation servos per finger) for the Texas A&M Robotic Hand Challenge.
- Engineered a closed-loop PID control architecture using AS5600 magnetic encoders read over I<sup>2</sup>C through a PCA9548A multiplexer for per-finger position feedback.
- Built a calibration workflow in MicroPython to capture and store named hand poses from live encoder data.
- Delivered the technical presentation that earned the team the Best Presentation Award and overall finalist standing.

#### FPGA Motor Speed Controller · Quartus Prime, DE10 FPGA, Verilog/VHDL, PWM

- Designed and implemented a 7-state, one-hot Moore state machine PWM motor speed controller on the DE10 FPGA.
- Drove a BCD 7-segment display output directly from FSM state logic.

#### STM32F4xx Embedded Systems Lab · C (bare metal), STM32F4xx, Keil uVision

- Programmed GPIO peripherals and SysTick timers at the register level in C with no HAL abstraction.

### EDUCATION

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- B.S., Electrical Systems Engineering Technology (in progress) · Texas A&M University, College Station, TX · Expected Dec. 2028
- Regents Scholar

### LEADERSHIP & INVOLVEMENT

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- Cohort 2 Member, KCLI Brotherhood · College Station, TX · Sept. 2024 – Present